

Mi Joo Jeong

<https://mdoublej2312.github.io/>

INTERESTS

Multimodal Vision-Language, Generative Vision, AR / VR, 3D Reconstruction & SLAM

EDUCATION

Sungkyunkwan University (SKKU)

B.S. in Computer Science and Engineering (Double Major)

B.S. in Architecture

Suwon, South Korea

Mar. 2020 – Aug. 2026

Mar. 2020 – Aug. 2026

RESEARCH AND WORK EXPERIENCE

HANKYEOL Architectural Firm

Student Internships

Seoul, South Korea

Jan. 2026 – Feb. 2026

- Gained hands-on experience in model making, façade design, and CAD drafting, and investigated housing project permit processes

Computer Vision Lab, SKKU

Undergraduate Research Assistant (Advisor: Prof. JuneHo Yi)

Suwon, South Korea

Dec. 2024 – Present

- Built a multimodal vision-language framework combining architectural floor plans and panoramic imagery to enhance spatial reasoning and design understanding
- Integrated generative vision and layered image decomposition to simulate dynamic lighting, reflection, and spatial experience in architectural visualization
- Applied SLAM-based 3D scene abstraction and multimodal scene generation to reconstruct and interact with built environments in immersive AR/VR architectural design

SecAI Lab, SKKU

Teaching Assistant (Advisor: Prof. HyungJoon Koo)

Suwon, South Korea

Dec. 2023 – Feb. 2024

- Created lecture slides and programming assignments on topics such as clickjacking, CSRF, Shellshock, SQL Injection, and XSS in web security and Heartbleed in network security, for Information Security Labs (previously known as Hands-on Security)

Learning and Adaptation Algorithms Lab, SNU

Undergraduate Research Assistant (Advisor: Prof. TaeSup Kim)

Seoul, South Korea

Jun. 2023 – Aug. 2023

- Reproduced and evaluated prompt-based adaptation frameworks (CoCoOp, ProDA, MaPLe, and PromptSRC), analyzing trade-offs between fine-tuning stability and zero-shot generalization
- Implemented and assessed advanced prompt-learning methods such as prompt distribution learning and self-regulating prompts to enhance robustness while preventing catastrophic forgetting.

PUBLICATIONS

[1] **Mijoo Jeong** and Wooyoung Kim, “AI-Based Approach for Predicting Congestion Levels in Train Station Spaces” (Ongoing)

[2] Sungho Kang, Hyunkyu Park, Yeonho Lee, Hanbyul Lee, **Mijoo Jeong**, YeongHyeon Park, Injae Lee, and Juneho Yi, “Anomaly Detection by Effectively Leveraging Synthetic Images,” In Proceedings of the IEEE 8th International Conference on Knowledge Innovation and Invention (ICKII), 2025

- Best Paper Award

PROJECTS

Computer Science and Engineering

Suwon, South Korea

AI Vision System for Wireless Charging Safety (Hyundai Motor Company)

Oct. 2025 – Present

- Developed an AI-based RGB + IR vision system for detecting foreign objects, living beings, and fire on wireless charging pads, including camera placement optimization and real-world edge inference testing

Education: Chemistry Basic Experiment through AR

Sep. 2025 – Dec. 2025

- Computer Science and Engineering Graduation Project

- Developed an AR marker-based chemistry experiment platform with teacher evaluation features and interactive student experiment content

Deepfake Generation and Detection

Jun. 2025 – Aug. 2025

- Built an end-to-end deepfake pipeline by generating synthetic faces with ProGAN and DDPM/DDIM and training a CNN-based detector for cross-dataset identification

SKKU Restaurant

Sep. 2024 – Dec. 2024

- Implemented the frontend in Kotlin with AWS Lambda backend integration, featuring Kakao Map API-based restaurant search and review summarization using Claude API

Hands-on-security

Dec. 2023 – Feb. 2024

- Developed web security lab exercises by creating or modifying HTML, CSS, and JavaScript files to demonstrate real-world attacks and defenses, including implementing the Heartbleed vulnerability to analyze leaked memory and automate sensitive data extraction

Architecture

Suwon, South Korea

Still-flow

Mar. 2025 – Jun. 2025

- Architecture Graduation Project
- Multi-cultural platform at Suseo Station for flow and stay, proposing a new model for future transportation hubs

All-around

Sep. 2024 – Dec. 2024

- Housing project reflecting the lifestyle of single residents and DINK couples without children

OTT (SeongSu)

Mar. 2024 – Jun. 2024

- Experience center in Seongsu-dong, a rising hotspot with increasing foreign visitors, designed for interactive engagement with OTT programs

Grab-ity

Aug. 2023 – Dec. 2023

- Remodeling project of Nam-Seoul Museum of Art in Sadang, creating diverse programs to attract frequent visits and active use

Free Strengthen

Feb. 2023 – Jun. 2023

- Office project designed for IT developers, providing a space to generate ideas and take rest from overwork

ACTIVITIES

NPC (Software Algorithm Club)

Club Member

Suwon, South Korea

Mar. 2024 – Dec. 2024

Comit (Coding Club)

Club Member

Suwon, South Korea

Mar. 2022 -Feb. 2023

Letz dance (Dance Club)

Leader

Seoul, South Korea

Mar. 2022 -Jul. 2022

TEACHING

High school mathematics tutoring

Teacher

Seoul, South Korea

Mar. 2021 – Sep. 2022

Seoul mathematics tutoring volunteer

Teaching assistant

Seoul, South Korea

Dec. 2021 – Feb. 2022

Seoul mathematics academy teaching

Teaching assistant

Seoul, South Korea

Mar. 2021 – Jun. 2021

Pulja online platform mathematics tutoring

Teacher

Seoul, South Korea

Mar. 2021 – Jun. 2021

TECHNICAL SKILLS

Programming Ability: Python, C/C++, Java, Html, Css, Javascript, PyTorch, Git, Shell, Docker, React

Architecture Ability: Adobe Photoshop, Adobe Illustrator, SketchUp, Rhino, AutoCAD, Enscape

Languages: Toefl 95 (Rc 24 Lc 23 Spk 20 Wrt 28), Chinese character proficient test grade 4